

Crime

pau, fer, y teo

Trabajo

Instalamos la bd

```
#install.packages("pder")  
library(pder)
```

Cargando paquete requerido: plm

```
data("Mafia")
```

limpiamos la bd

```
# Cargar librerías  
library(sf)
```

Linking to GEOS 3.14.1, GDAL 3.12.2, PROJ 9.7.1; sf_use_s2() is TRUE

```
library(dplyr)
```

Adjuntando el paquete: 'dplyr'

The following objects are masked from 'package:plm':

between, lag, lead

The following objects are masked from 'package:stats':

filter, lag

The following objects are masked from 'package:base':

intersect, setdiff, setequal, union

```
# Cargar shapefile de Italia
italy <- st_read("it_shp/it.shp", quiet = TRUE)

# Tabla de equivalencias entre nombres del shapefile y del dataset Mafia
equivalencias <- data.frame(
  shp_name = c("Turin", "Aoste", "Mantova", "Bozen", "Padova",
               "Reggio Emilia", "Forlì-Cesena", "Massa-Carrara",
               "Firenze", "Pesaro e Urbino", "Reggio Calabria",
               "Siracusa", "Oristrano"),
  mafia_name = c("Torino", "Aosta", "Mantua", "Bolzano", "Padua",
                 "Reggio Nell'Emilia", "Forli' - Cesena", "Massa Carrara",
                 "Florence", "Pesaro E Urbino", "Reggio Di Calabria",
                 "Syracuse", "Oristano")
)

# Filtrar y renombrar provincias para que coincidan con Mafia
italy_clean <- italy |>
  left_join(equivalencias, by = c("name" = "shp_name")) |>
  mutate(province = ifelse(is.na(mafia_name), name, mafia_name)) |>
  filter(province %in% unique(Mafia$province))
plot(st_geometry(italy_clean))
```



matriz de pesos

```
library(spdep)
```

Cargando paquete requerido: spData

To access larger datasets in this package, install the spDataLarge package with: ``install.packages('spDataLarge', repos='https://nowosad.github.io/drat/', type='source')``

```
library(sf)
```

```
# Centroides y matriz W por knn=4  
coords  <- st_centroid(italy_clean) |> st_coordinates()
```

Warning: st_centroid assumes attributes are constant over geometries

```
italy_nb <- knn2nb(knearneigh(coords, k = 4))  
italy_W  <- nb2listw(italy_nb, style = "W")
```